

Questions with Answer Keys

MathonGo

Q1 - 2024 (01 Feb Shift 1)

If R is the radius of the earth and the acceleration due to gravity on the surface of earth is $g = \pi^2 \text{ m/s}^2$, then the length of the second's pendulum at a height $h = 2R$ from the surface of earth will be,:

(1) $\frac{2}{9} \text{ m}$

(2) $\frac{1}{9} \text{ m}$

(3) $\frac{4}{9} \text{ m}$

(4) $\frac{8}{9} \text{ m}$

Q2 - 2024 (01 Feb Shift 2)

A light planet is revolving around a massive star in a circular orbit of radius R with a period of revolution T . If the force of attraction between planet and star is proportional to $R^{-3/2}$ then choose the correct option :

(1) $T^2 \propto R^{5/2}$

(2) $T^2 \propto R^{7/2}$

(3) $T^2 \propto R^{3/2}$

(4) $T^2 \propto R^3$

Q3 - 2024 (27 Jan Shift 1)

The acceleration due to gravity on the surface of earth is g . If the diameter of earth reduces to half of its original value and mass remains constant, then acceleration due to gravity on the surface of earth would be :

(1) $g/4$

(2) $2g$

(3) $g/2$

(4) $4g$

Q4 - 2024 (27 Jan Shift 2)

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The angular speed of the moon in its orbit about the earth is more than the angular speed of the

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Questions with Answer Keys

MathonGo

earth in its orbit about the sun.

Reason (R): The moon takes less time to move around the earth than the time taken by the earth to move around the sun.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) (A) is correct but (R) is not correct
- (2) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (3) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (4) (A) is not correct but (R) is correct

Q5 - 2024 (29 Jan Shift 1)

At what distance above and below the surface of the earth a body will have same weight, (take radius of earth as R .)

- (1) $\sqrt{5}R - R$
- (2) $\frac{\sqrt{3}R - R}{2}$
- (3) $\frac{R}{2}$
- (4) $\frac{\sqrt{5}R - R}{2}$

Q6 - 2024 (29 Jan Shift 2)

A planet takes 200 days to complete one revolution around the Sun. If the distance of the planet from Sun is reduced to one fourth of the original distance, how many days will it take to complete one revolution?

- (1) 25
- (2) 50
- (3) 100
- (4) 20

Q7 - 2024 (30 Jan Shift 1)

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Questions with Answer Keys

MathonGo

The gravitational potential at a point above the surface of earth is $-5.12 \times 10^7 \text{ J/kg}$ and the acceleration due to gravity at that point is 6.4 m/s^2 . Assume that the mean radius of earth to be 6400 km. The height of this point above the earth's surface is :

- (1) 1600 km
- (2) 540 km
- (3) 1200 km
- (4) 1000 km

Q8 - 2024 (30 Jan Shift 2)

Escape velocity of a body from earth is 11.2 km/s . If the radius of a planet be one-third the radius of earth and mass be one-sixth that of earth, the escape velocity from the planet is:

- (1) 11.2 km/s
- (2) 8.4 km/s
- (3) 4.2 km/s
- (4) 7.9 km/s

Q9 - 2024 (31 Jan Shift 1)

Four identical particles of mass m are kept at the four corners of a square. If the gravitational force exerted on one of the masses by the other masses is $\left(\frac{2\sqrt{2}+1}{32}\right) \frac{Gm^2}{L^2}$, the length of the sides of the square is

- (1) $\frac{L}{2}$
- (2) $4L$
- (3) $3L$
- (4) $2L$

Q10 - 2024 (31 Jan Shift 2)

The mass of the moon is $1/144$ times the mass of a planet and its diameter $1/16$ times the diameter of a planet. If the escape velocity on the planet is v , the escape velocity on the moon will be:

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Questions with Answer Keys

MathonGo

(1) $\frac{v}{3}$						
(2) $\frac{v}{4}$						
(3) $\frac{v}{12}$						
(4) $\frac{v}{6}$						

Questions with Answer Keys

MathonGo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

Answer Key

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

Q1 (2) mathongo /// ma**Q2** (1) mathongo **Q3** (4) mathongo /// ma**Q4** (2) mathongo

Q5 (4) mathongo /// ma**Q6** (1) mathongo **Q7** (1) mathongo /// ma**Q8** (4) mathongo

Q9 (2) mathongo /// ma**Q10** (1) mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

/// mathongo /// mathongo /// mathongo /// mathongo /// mathongo /// mathongo

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Solutions

MathonGo

Q1

$$g' = \frac{GM_e}{(3R)^2} = \frac{1}{9} g$$

$$T = 2\pi \sqrt{\frac{\ell}{g'}}$$

Since the time period of second pendulum is 2sec.

$$T = 2\text{sec}$$

$$2 = 2\pi \sqrt{\frac{\ell}{g}}$$

$$\ell = \frac{1}{9} \text{ m}$$

Q2

$$F = \frac{GMm}{R^{3/2}} = m\omega^2 R$$

$$\omega^2 \propto \frac{1}{R^{5/2}} \therefore T = \frac{2\pi}{\omega} \text{ so}$$

$$T^2 \propto R^{5/2}$$

Q3

$$g = \frac{GM}{R^2} \Rightarrow g \propto \frac{1}{R^2}$$

$$\frac{g_2}{g_1} = \frac{R_1^2}{R_2^2}$$

$$g_2 = 4 g_1 \left(R_2 = \frac{R_1}{2} \right)$$

Q4

$$\omega = \frac{2\pi}{T} \Rightarrow \omega \propto \frac{1}{T}$$

$$T_{\text{moon}} = 27 \text{ days}$$

$$T_{\text{earth}} = 365 \text{ days } 4 \text{ hour}$$

$$\Rightarrow \omega_{\text{moon}} > \omega_{\text{earth}}$$

Q5

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Solutions

MathonGo



$$g_p = \frac{gR^2}{(R+h)^2}$$

$$g_q = g \left(1 - \frac{h}{R} \right)$$

$$g_p = g_q$$

$$\frac{g}{\left(1 + \frac{h}{R} \right)^2} = g \left(1 - \frac{h}{R} \right)$$

$$\left(1 - \frac{h^2}{R^2} \right) \left(1 + \frac{h}{R} \right) = 1$$

$$\text{Take } \frac{h}{R} = x$$

So

$$x^3 - x + x^2 = 0$$

$$x = \frac{\sqrt{5}-1}{2}$$

$$h = \frac{R}{2}(\sqrt{5}-1)$$

Q6

$$T^2 \propto r^3$$

$$\frac{T_1^2}{r_1^3} = \frac{T_2^2}{r_2^3}$$

$$\frac{(200)^2}{r^3} = \frac{T_2^2}{\left(\frac{r}{4} \right)^3}$$

$$\frac{200 \times 200}{4 \times 4 \times 4} = T_2^2$$

$$T_2 = \frac{200}{4 \times 2}$$

$$T_2 = 25 \text{ days}$$

Q7

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Solutions

MathonGo

$$-\frac{GM_E}{R_E+h} = -5.12 \times 10^{-7} \dots (i)$$

$$\frac{GM_E}{(R_E+h)^2} = 6.4 \dots (ii)$$

By (i) and (ii)

$$\Rightarrow h = 16 \times 10^5 \text{ m} = 1600 \text{ km}$$

Q8

$$R_P = \frac{R_E}{3}, M_P = \frac{M_E}{6}$$

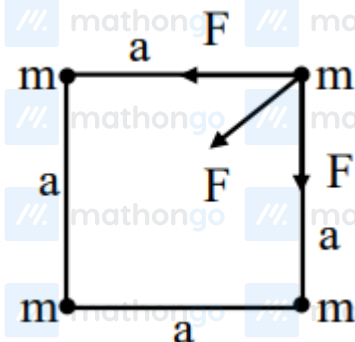
$$V_e = \sqrt{\frac{2GM_e}{R_e}} \dots (i)$$

$$V_P = \sqrt{\frac{2GM_P}{R_P}} \dots (ii)$$

$$\frac{V_e}{V_P} = \sqrt{2}$$

$$V_P = \frac{V_e}{\sqrt{2}} = \frac{11.2}{\sqrt{2}} = 7.9 \text{ km/sec}$$

Q9



$$F_{\text{net}} = \sqrt{2}F + F'$$

$$F = \frac{Gm^2}{a^2} \text{ and } F' = \frac{Gm^2}{(\sqrt{2}a)^2}$$

$$F_{\text{net}} = \sqrt{2} \frac{Gm^2}{a^2} + \frac{Gm^2}{2a^2}$$

$$\left(\frac{2\sqrt{2}+1}{32} \right) \frac{Gm^2}{L^2} = \frac{G^2}{a^2} \left(\frac{2\sqrt{2}+1}{2} \right)$$

$$a = 4L$$

Q10

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Solutions

MathonGo

$$V_{\text{escape}} = \sqrt{\frac{2GM}{R}}$$

$$V_{\text{planet}} = \sqrt{\frac{2GM}{R}} = V$$

$$V_{\text{Moon}} = \sqrt{\frac{2GM \times 16}{144R}} = \frac{1}{3} \sqrt{\frac{2GM}{R}}$$

$$V_{\text{Moon}} = \frac{V_{\text{Planet}}}{3} = \frac{V}{3}$$

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)